

The Crossing Number of the Hypercube

OPG-313

L1 - TRACTABLE

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Graph Theory

ORIGINAL PROBLEM STATEMENT

The crossing number $cr(G)$ of G is the minimum number of crossings in all drawings of G in the plane.

The d -dimensional (hyper)cube Q_d is the graph whose vertices are all binary sequences of length d , and two of the sequences are adjacent in Q_d if they differ in precisely one coordinate.

Conjecture $\lim_{d \rightarrow \infty} \frac{cr(Q_d)}{4^d} = \frac{5}{32}$

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Archive: <https://www.unsolvedmath.com/problems?difficulty=1>

Dataset: <https://huggingface.co/datasets/ulamai/UnsolvedMath>